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Start of meeting _____

1 Progress and Pre-Meeting Notes _____

Recent experimental results were presented and reviewed, focusing on the comparative performance between the LSTM and TFT architectures under different configuration parameters.

2 Experimental Results and Observations _____

2.1 Architecture Performance Comparison

- **LSTM Superiority on Segregated Passes:** Initial benchmarks indicate that the LSTM model currently outperforms the Temporal Fusion Transformer (TFT) when the dataset is filtered and trained on separate pass types (i.e., Pre-Finishing and Finishing handled independently).
- **Anomalous Cross-Validation Fold:** A distinct divergence was observed in one specific cross-validation fold. This anomaly is driven by Tools 9, 10, and 11, which exhibit highly atypical physical profiles compared to the rest of the dataset.

3 Next Steps and Future Benchmarks _____

3.1 Target and Windowing Matrix Configuration

To fully evaluate the models, a systematic benchmark matrix will be executed across the following four experimental axes, segregated by pass type:

- **1 Line per File (Global Aggregation):**
 - Predict the **Mean** torque value ($y = \text{Mean Torque}$).
 - Predict the **Median** torque value ($y = \text{Median Torque}$).
- **10 Lines per File (Windowed Segments):**

- 20 ▶ Predict the **Mean** torque value ($y = \text{Mean Torque}$).
 ▶ Predict the **Median** torque value ($y = \text{Median Torque}$).

4 Repository and Code Quality

4.1 Refactoring and Documentation

- 25 • **Project Structure Cleanup:** Immediate priority is given to cleaning up the code architecture and directories to ensure the pipeline is clean, self-contained, and highly intuitive for external review.
- **Documentation:** The repository `README.md` must be thoroughly updated to document the current v6 pipeline execution, dataset variants, and validation splits.

Meeting Adjourned

30 *Notes reformulated with AI assistance*